Smart Credit Card Approval System

AI-Based Loan Eligibility Prediction System Documentation

Date: 30 June 2026
Track: AI-ML and Gen AI Track

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1. System Overview

Smart Credit Card Approval System is an AI-powered application that helps banks and financial institutions evaluate **credit card applications** quickly and accurately. The application integrates a **Machine Learning model**, a **Flask-based web application**, and a **MySQL database** to analyze applicant information such as age, annual income, employment status, credit score, existing loans, repayment history, debt-to-income ratio, and financial behavior.

The system predicts whether an applicant is eligible for a **credit card** and provides an **approval probability**. It reduces manual verification time, improves consistency in approval decisions, minimizes financial risk, and enhances customer experience through a secure and user-friendly web interface.

2. System Architecture

The application follows a layered architecture consisting of:

- **Presentation Layer** – HTML5, CSS3, JavaScript, Bootstrap
- **Application Layer** – Flask REST API
- **Business Logic Layer** – Authentication, Data Validation, Credit Card Application Processing
- **Machine Learning Layer** – Scikit-learn Credit Card Approval Prediction Model
- **Data Layer** – MySQL Database using SQLAlchemy ORM

POST /api/predict
Receives applicant information, validates the input data, predicts **credit card approval** using the trained Machine Learning model, stores the prediction, and returns the approval result.

Request Format:-

```
{
  "age": 28,
  "gender": "Male",
  "employment_status": "Salaried",
  "annual_income": 650000,
  "credit_score": 780,
  "existing_loans": 1,
  "monthly_expenses": 25000,
  "debt_to_income_ratio": 0.32,
  "marital_status": "Single"
}
```

Response Format:-

```
{
  "credit_card_status": "Approved",
  "approval_probability": "96.8%"
}
```

3. Setup & Installation Instructions

Follow these steps to set up and run the **Smart Credit Card Approval System** locally.

Step 1 – Create Virtual Environment

```
python -m venv venv
```

Step 2 – Activate Virtual Environment

```
venv\Scripts\activate
```

Step 3 – Install Dependencies

```
pip install -r requirements.txt
```

Step 4 – Configure Database Create a MySQL database named:

```
smart_credit_card
```

Update the database credentials in **config.py**.

Step 5 – Train the Machine Learning Model

```
python Training/train_model.py
```

```
The trained model will be saved as:
 credit_card_model.pkl
```

Step 6 – Run the Flask Application

```
python app.py
```

Step 7 – Open the Application

```
http://127.0.0.1:5000
```